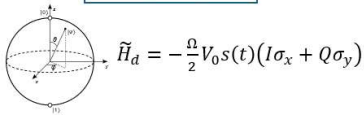


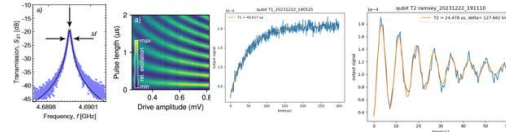
Last Lecture

RECALLING

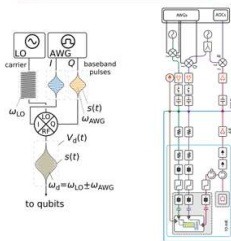
Single qubit gates



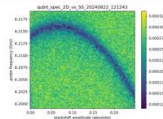
Qubit characterization step-by-step



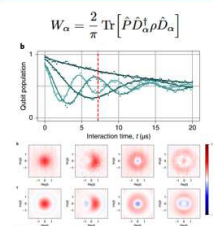
Experimental setup



Selection of interaction type



Phase space representation



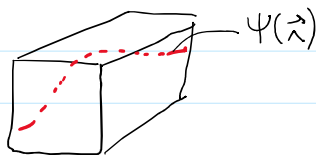
$$I = I_c \sin \varphi$$

Today's Lecture : Superconductivity

Superconductivity is a phenomenon related to many-bodies physics. To understand it, let's understand some basic concepts.

We have already discussed the role of a wavefunction. For instance, we use its amplitude square to describe the probability of finding a particle at a given position

Ex:



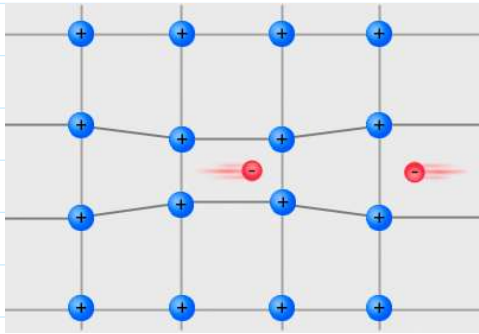
A particle in a box can be found at $\vec{r}$ with probability $\Psi \cdot \Psi^*$

If the particle is a fermion, such as an electron, a second particle can not occupy the same state (exclusion principle of Pauli). However, if we have bosons, many of them can occupy the same state.

We observe in many metals that when we cool them down

We observe in many metals that when we cool them down below a certain temperature, they become superconducting. There is a theory, called BCS theory, that explains the phase transition. For our discussions, it will be enough a phenomenological description of it.

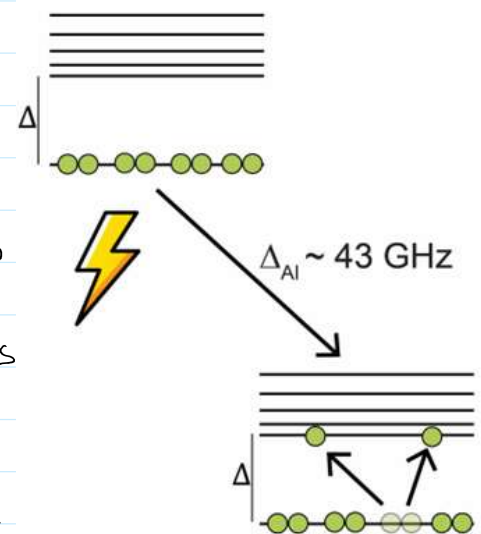
Cooper Pairs



It turns out that, at very low temperatures, the electrons interact with vibrations of the lattice creating an effective attraction between them. Pairs of interacting electrons are called Cooper pairs.

A bound pair acts a bosonic particle (2 spin- $\frac{1}{2}$ particles), and several of these pairs can occupy the same state. In fact, we can assume that a single wavefunction will be enough to describe many of these pairs.

The Cooper pairs effectively condense to a single state, with a binding energy 2Δ between the pairs. This can be described using an energy diagram, with a bandgap Δ . If energy enough is given to the system, we can break Cooper pairs, creating the so-called quasiparticles, electrons uncoupled traveling in the lattice. These are an important source of decay for SC qubits. The energy gap Δ is a material and temperature dependent value. At $T=0$, it is related to the critical temperature T_c (the phase transition temperature) as:



$$\Delta(T=0) \cong 1.764 k_B T_c \quad (\text{BCS theory})$$

List [edit]

Substance	Class	T_C (K)
Al	Element	1.20
Bi	Element	5.3×10^{-4}
Cd	Element	0.52
Diamond:B	Element	11.4
Ga	Element	1.083
Hf	Element	0.165
α -Hg	Element	4.15
β -Hg	Element	3.95
In	Element	3.4
Ir	Element	0.14
α -La	Element	4.9
β -La	Element	6.3
Li	Element	4×10^{-4}
Mo	Element	0.92
Nb	Element	9.26
Os	Element	0.65
Pa	Element	1.4
Pb	Element	7.19
Re	Element	2.4
Rh	Element	3.25×10^{-4}
Ru	Element	0.49
Si:B	Element	0.4
Sn	Element	3.72
Ta	Element	4.48
Tc	Element	7.46–11.2

Considering Q8, the higher T_C the better

Order parameter

Before, we discussed that the wavefunction described the probability of finding a single particle in a specific position. Now, with many particles, we can associate the square of the wavefunction with a density of particles $n(\vec{r}, t)$

$$|\Psi(\vec{r}, t)|^2 = n(\vec{r}, t)$$

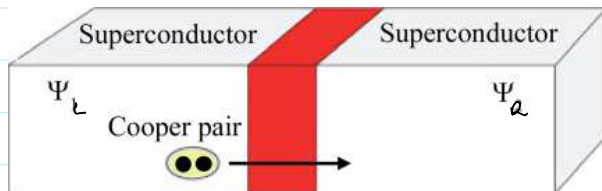
This wavefunction will also carry a specific phase. So, a reasonable ansatz for its form is

$$\Psi(\vec{r}, t) = \sqrt{n(\vec{r}, t)} e^{i\theta(\vec{r}, t)}$$

The system of many particles (Cooper pairs) will behave as a condensate with this order parameter

Josephson Junctions

Assume now that we have two SC islands separated by an insulator



Each side will have its order parameter $\Psi_{R,L} = \sqrt{n_{R,L}} e^{i\theta_{R,L}}$

The dynamics of the systems will be given by Schrödinger-like equations with a coupling term, i.e.,

$$i\hbar \frac{\partial \Psi_L}{\partial t} = \mu_L \Psi_L - K \Psi_R \quad i\hbar \frac{\partial \Psi_R}{\partial t} = \mu_R \Psi_R - K \Psi_L$$

(They are in fact mean field equations for the order parameter)

Using our developed order parameters, we have for Ψ_L

$$i\hbar \left(\frac{\dot{n}_L}{2n_L} + i\dot{\theta}_L \right) \Psi_L = \mu_L \Psi_L - K \Psi_R$$

Dividing by Ψ_L

$$i\hbar \left(\frac{\dot{n}_L}{2n_L} + i\dot{\theta}_L \right) = \mu_L - K \frac{\Psi_R}{\Psi_L} \quad (1)$$

and $\frac{\Psi_R}{\Psi_L} = \sqrt{\frac{n_R}{n_L}} e^{i(\theta_R - \theta_L)}$. Defining $\varphi = \theta_L - \theta_R$

$$\frac{\Psi_R}{\Psi_L} = \sqrt{\frac{n_R}{n_L}} e^{-i\varphi} = \sqrt{\frac{n_R}{n_L}} (\cos\varphi - i\sin\varphi)$$

Now, we can split the imaginary and real parts of (1)

• Imaginary part

$$\frac{\hbar}{2} \frac{\dot{n}_L}{n_L} = K \sqrt{\frac{n_R}{n_L}} \sin \varphi \rightarrow \dot{n}_L = \frac{2K}{\hbar} \sqrt{n_R n_L} \sin \varphi$$

Therefore, we found a flux of Cooper pairs. This can be translated to a current as

$$I = 2e \dot{n} = \underbrace{\frac{4eK}{\hbar} \sqrt{n_R n_L}}_{\equiv I_c} \sin \varphi$$

$$I = I_c \sin \varphi$$

This is the first Josephson relation, used in our derivations of a qubit Hamiltonian.

• Real part

We use both equations

$$-\hbar \dot{\theta}_L = \mu_L - K \sqrt{\frac{n_R}{n_L}} \cos \varphi \quad -\hbar \dot{\theta}_R = \mu_R - K \sqrt{\frac{n_L}{n_R}} \cos \varphi$$

Subtracting one from the other, we have

$$-\hbar \underbrace{(\dot{\theta}_L - \dot{\theta}_R)}_{\dot{\varphi}} = \mu_L - \mu_R - K \left(\sqrt{\frac{n_R}{n_L}} - \sqrt{\frac{n_L}{n_R}} \right) \cos \varphi$$

Assuming $n_R \approx n_L$

$$-\hbar \dot{\varphi} = \mu_L - \mu_R$$

this difference in energy across the junction can come from a voltage drop.

a voltage drop.

$$\text{So } \hbar \dot{\varphi} = 2eV \rightarrow \dot{\varphi} = \frac{2e}{\hbar} V$$

defining $\Phi_0 = \frac{h}{2e}$ as the flux quantum

$$V = \frac{\Phi_0}{2\pi} \frac{d\varphi}{dt}$$

This is the second Josephson relation.

Consequences

- If $V=0$, $\dot{\varphi}=0 \rightarrow$ Static phase difference
 $\rightarrow$ static current
(DC Josephson effect)
- If $V \neq 0$, $\varphi(t) = \varphi_0 + \frac{2e}{\hbar} Vt \rightarrow$ supercurrent oscillates
(AC Josephson effect)
- If we define $\Phi = \frac{\Phi_0}{2\pi} \varphi$, we have

$$V = \frac{d\Phi}{dt}$$

which is equivalent to Faraday's law of induction but, instead of a magnetic field, this voltage comes from the kinetic energy of Cooper pairs. This effect is known as kinetic inductance

Connecting with qubits

We used JJ as nonlinear inductors. Let's see how this shows up. We use

$$\frac{\partial I}{\partial \phi} = I_c \cos \phi$$

$$\frac{\partial \phi}{\partial t} = \frac{2\pi}{\Phi_0} V$$

So

$$\frac{\partial I}{\partial t} = \frac{\partial I}{\partial \phi} \frac{\partial \phi}{\partial t} = I_c \cos \phi \frac{2\pi}{\Phi_0} V$$

which rearranging leads to

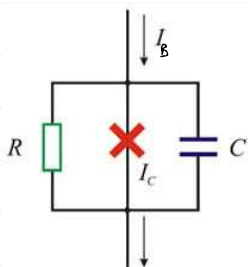
$$V = \frac{\Phi_0}{2\pi I_c \cos \phi} \frac{\partial I}{\partial t} = L_J \frac{\partial I}{\partial t}$$

and we found the previously used non-linear inductance

$$L_J = \frac{\Phi_0}{2\pi I_c \cos \phi}$$

RCSJ model

In fact, when introducing a JJ in a circuit, we have to consider that it has a resistance (given by the quasiparticles) and a capacitance of the pads. This leads to the resistively and capacitively shunted junction model



We have that

$$I_B = I_c \sin \phi + \frac{V}{R} + C \dot{V}$$

Replacing $V = \frac{\hbar}{2e} \dot{\phi}$

$$I_B = I_c \sin \phi + \frac{\hbar}{2e R} \dot{\phi}^2 + C \hbar \ddot{\phi}$$

$$I_B = I_c \sin \phi + \frac{\hbar}{2eR} \dot{\phi} + \frac{C\hbar}{2e} \ddot{\phi}$$

or, Rearranging

$$\frac{C\hbar}{2e} \ddot{\phi} + \frac{\hbar}{2eR} \dot{\phi} + I_c \sin \phi - I_B = 0$$

Multiplying by $\frac{\hbar}{2e}$, we define

$$\underbrace{M \ddot{\phi}}_{\text{inertia}} + \underbrace{\eta \dot{\phi}}_{\text{viscous damping}} + \underbrace{\frac{\partial U}{\partial \phi}}_{\text{conservative force}} = 0$$

$$M \equiv C \left(\frac{\hbar}{2e} \right)^2$$

$$\eta \equiv \frac{1}{R} \left(\frac{\hbar}{2e} \right)^2$$

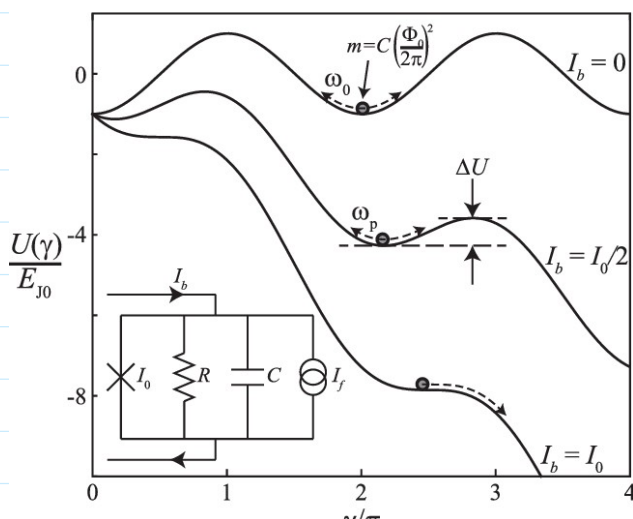
And

$$\frac{\partial U}{\partial \phi} = E_J \sin \phi - \frac{\hbar}{2e} I_B$$

$$E_J \equiv \frac{\hbar I_c}{2e}$$

$$\rightarrow U(\phi) = -E_J \cos \phi - \frac{\hbar}{2e} I_B \phi$$

If we plot this, we see that it is a "tilted washboard" potential

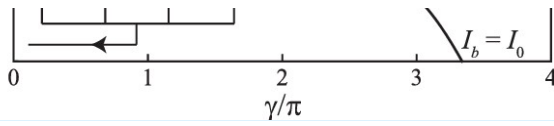


If we treat ϕ as a "particle", it is trapped in a valley if

$$\langle \dot{\phi} \rangle = 0 \rightarrow \langle \phi \rangle = 0 \text{ (DC)}$$

If the tilt is large enough, the phase runs

$$\phi \rightarrow \phi + 2\pi$$



energy, in phase with

finite $V \rightarrow \langle \dot{\varphi} \rangle > 0$

Ambeghakar-Baratoff formula

There is a simple way to measure the critical current
At room temperature, with AB formula

$$I_c \cong \frac{\pi \Delta_0}{2e R_n}$$

$\Delta_0 \rightarrow$ material bandgap

$R_n \rightarrow$ normal state resistance